

Notes on lecture 15

Jiten Chandra Kalita and Shyamashree Upadhyay

1 Line integrals

Recall that in single variable calculus, for integrating a continuous function $f(x)$ between the points a and b , we had divided the interval $[a, b]$ into subintervals $[x_{i-1}, x_i]$ ($1 \leq i \leq n$), where

$$a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b.$$

Then in the interval $[x_{i-1}, x_i]$, we had taken a point x_i^* and we had considered the limit $\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x_i$ of the Riemann sum, where $\Delta x_i = x_i - x_{i-1}$. This limit was defined as $\int_a^b f(x) dx$. This is an example of a line integral over the straight line which is the interval $[a, b]$ of the real line.

1.1 Line integrals along plane curves

Now we are going to consider a different type of line integral. Let us first consider a plane curve C (in the xy -plane). Assume that this curve C can be parametrized by a single variable t in such a way that if $a \leq t \leq b$, then the curve C can be defined by points like (x, y) where x and y are functions of t . That is, $x = x(t), y = y(t)$. We are also going to assume that C is a continuous curve and C is smooth (By smooth, we mean that at each point $P = (x(t), y(t))$ of the curve, if we consider the position vector and call it $\vec{r}(t)$, then $\vec{r}'(t) \neq \vec{0}$. In other words, we can draw a tangent at each point of the curve.).

Now we divide the interval $[a, b]$ (in which the parameter t lies) into subintervals $[t_{i-1}, t_i]$ ($1 \leq i \leq n$) where

$$a = t_0 < t_1 < \cdots < t_{n-1} < t_n = b.$$

For each $i \in \{1, \dots, n\}$, we are going to get two points P_{i-1} and P_i on the curve, whose coordinates are $(x(t_{i-1}), y(t_{i-1}))$ and $(x(t_i), y(t_i))$ respectively. Let us assume that the length of the subarc $P_{i-1}P_i$ (along the curve C) is Δs_i . We are going to take a point $P_i^* = (x_i^*, y_i^*)$ on this subarc. This means that we can find a point $t_i^* \in [t_{i-1}, t_i]$ such that $x_i^* = x(t_i^*)$ and $y_i^* = y(t_i^*)$. Please see Figure 1.1.1 for an illustration.

If f is a continuous function of two variables x and y defined in some domain

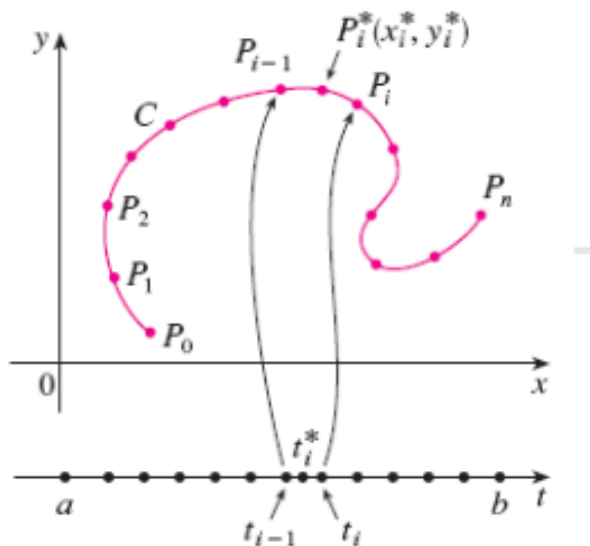


Figure 1.1.1: Line integral along a plane curve C

D , and if the curve C lies in the domain D , then we consider the Riemann sum

$$\sum_{i=1}^n f(x_i^*, y_i^*) \Delta s_i.$$

We define the **line integral** $\int_C f(x, y) ds$ **along the curve** C as the limit

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*, y_i^*) \Delta s_i.$$

Recall from the previous lectures that

$$ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

So we have

$$\int_C f(x, y) ds = \int_a^b f(x(t), y(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

The upper and lower limits of integration in the right hand side integral in the above equation are b and a respectively because the end points of the curve C correspond to $t = a$ and $t = b$. This finishes the definition of line integrals along a curve C in the xy -plane.

Geometric interpretation: Please refer to Figure 1.1.2 below.

At each point (x, y) on the curve C in this figure, we consider the height

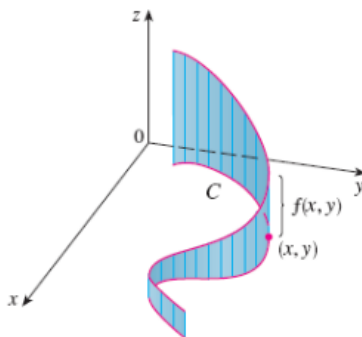


Figure 1.1.2: Geometric interpretation of line integral

$f(x, y)$. By computing the line integral of f along the curve C , we are simply finding out the surface area of the fence erected along the curve C (as shown in Figure 1.1.2), where at each point (x, y) of C , the height of the fence is $f(x, y)$.

Example 1.1.1. Evaluate $\int_C (2 + x^2 y) ds$, where C is the upper half of the circle $x^2 + y^2 = 1$. Figure 1.1.3 shows the curve C in the xy -plane.

Whenever we talk about a line integral along a curve, we will have to think

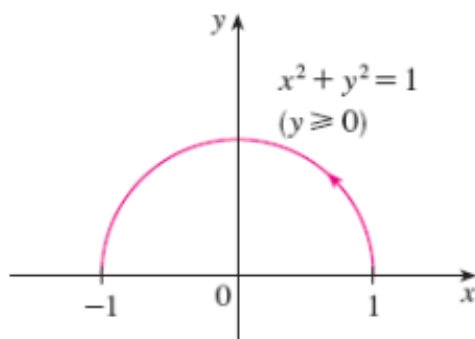


Figure 1.1.3: Figure for Example 2.1.1

about a parametrization of the curve. Here, the curve C can be parametrized by

$$x = \cos t, \quad y = \sin t, \quad 0 \leq t \leq \pi.$$

So we have

$$\frac{dx}{dt} = -\sin t \quad \text{and} \quad \frac{dy}{dt} = \cos t.$$

Then

$$\begin{aligned} \int_C (2 + x^2 y) ds &= \int_0^\pi (2 + \cos^2 t \sin t) dt \\ &= \left[2t - \frac{\cos^3 t}{3} \right]_0^\pi. \end{aligned}$$

The rest of the calculation is left to the reader as an exercise. $\square$

1.2 Line integrals along piecewise smooth curves

Earlier we had assumed that C is a smooth curve. But now, we may assume that the curve C is piecewise smooth, that is, smooth along certain portions. Figure 1.2.1 shows a piecewise smooth curve.

Say, $C = C_1 \cup C_2 \cup \cdots \cup C_n$, where each C_i is a smooth curve. Then

$$\int_C f(x, y) ds = \int_{C_1} f(x, y) ds + \cdots + \int_{C_n} f(x, y) ds.$$

Remark 1.2.1. Whenever we are computing a line integral along a curve, we need not worry about the integrand. We only need to worry about the curve and its parametrization.

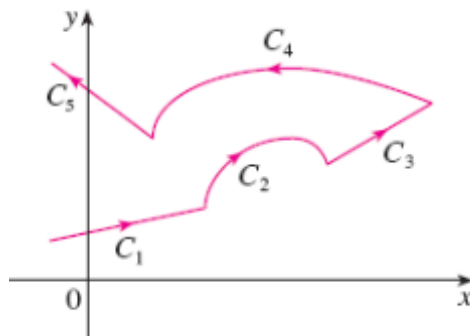


Figure 1.2.1: A piecewise smooth curve

Example 1.2.2. Evaluate $\int_C 2x ds$ where $C = C_1 \cup C_2$, C_1 is given by $y = x^2$ between the points $(0, 0)$ and $(1, 1)$, and C_2 is the vertical line between the points $(1, 1)$ and $(1, 2)$. Figure 1.2.2 shows the curve C .

Obviously,

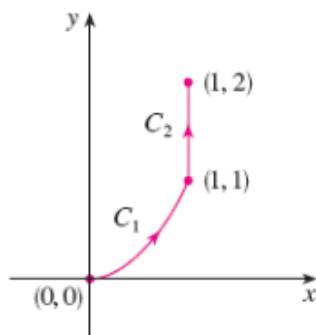


Figure 1.2.2: Figure for Example 2.2.2

$$\int_C 2x ds = \int_{C_1} 2x ds + \int_{C_2} 2x ds.$$

Observe that the curve C_1 can be simply parametrized by putting $x = t$ and $y = t^2$. Here x ranges from 0 to 1. So t will also range from 0 to 1. Then

$$\int_{C_1} 2x ds = \int_0^1 2t \sqrt{1 + 4t^2} dt.$$

Similarly, C_2 can be simply parametrized by $x = 1, y = t$ [Because along C_2 , x is fixed as 1.]. And t will vary from 1 to 2. Then

$$\int_{C_2} 2x ds = \int_1^2 2 \cdot 1 \sqrt{0+1} dt.$$

The rest of the calculation is left to the reader as an exercise. $\square$

1.3 Mass and center of mass of wire

Please read pages 13-14 of Lecture 15 for this.

1.4 Line integrals with respect to x and y

Earlier we had

$$\int_C f(x, y) ds = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*, y_i^*) \Delta s_i.$$

If instead of Δs_i , we put $\Delta x_i = x_i - x_{i-1}$ (where $x_i = x(t_i)$ and $x_{i-1} = x(t_{i-1})$), then we define

$$\int_C f(x, y) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*, y_i^*) \Delta x_i.$$

Similarly, we define

$$\int_C f(x, y) dy = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*, y_i^*) \Delta y_i,$$

where $\Delta y_i = y_i - y_{i-1}$, $y_i = y(t_i)$, and $y_{i-1} = y(t_{i-1})$.

Once we know a parametrization of the curve C (by the parameter t), we can simply write $\int_C f(x, y) dx$ as $\int_a^b f(x(t), y(t)) \frac{dx}{dt} dt$ or $\int_a^b f(x(t), y(t)) x'(t) dt$. Similarly for $\int_C f(x, y) dy$.

Example 1.4.1. Evaluate $\int_C y^2 dx + x dy$, where

- (a) $C = C_1$ is the line segment from $(-5, -3)$ to $(0, 2)$ and
 - (b) $C = C_2$ is the arc of the parabola $x = 4 - y^2$ from $(-5, -3)$ to $(0, 2)$.
- Please see Figure 1.4.1 for the graph of C .

Consider the integral $\int_{C_1} y^2 dx + x dy$. The equation of C_1 is given by

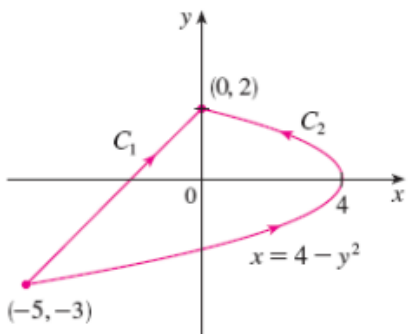


Figure 1.4.1: Figure for Example 2.4.1

$$y - 2 = \frac{-3 - 2}{-5 - 0}(x - 0)$$

or,

$$y = x + 2.$$

The easiest way to parametrize this curve C_1 is to take

$$x = t, y = t + 2$$

where t varies from -5 to 0 . Then

$$\begin{aligned} \int_{C_1} y^2 dx + x dy &= \int_{-5}^0 [(t+2)^2 dt + t dt] \\ &= \int_{-5}^0 (t^2 + 5t + 4) dt. \end{aligned}$$

Consider C_2 now. The easiest way to parametrize it is

$$y = t, x = 4 - t^2$$

where t varies from -3 to 2 . Then

$$\begin{aligned} \int_{C_2} y^2 dx + x dy &= \int_{-3}^2 [t^2(-2t dt) + (4 - t^2) dt] \\ &= \int_{-3}^2 (-2t^3 - t^2 + 4) dt. \end{aligned}$$

□

Remark 1.4.2. The integrals $\int_C f(x, y)dx$ and $\int_C f(x, y)dy$ are called line integrals with respect to x and y respectively. The line integral $\int_C f(x, y)ds$ is called line integral with respect to arc length. So whenever we are integrating with respect to s , the integration is going to be with respect to arc length.

1.5 Orientation

Consider a curve C (from the point A to B) and suppose it can be parametrized by the parameter t where $a \leq t \leq b$. For the point A , if the coordinates of A are (x, y) , then $x = x(a)$ and $y = y(a)$. Similarly, for the point B , if the coordinates of B are (x, y) , then $x = x(b)$ and $y = y(b)$. Let us assume that the orientation of the curve C is from A to B . Now, if we consider the same curve with opposite orientation (that is, from B to A), then we denote the curve by $-C$ (see Figure 1.5.1 for an illustration). Then we have

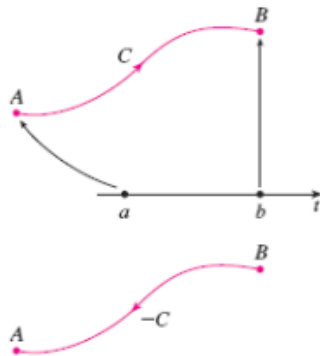


Figure 1.5.1: The curves C and $-C$

$$\int_{-C} f(x, y)ds = \int_C f(x, y)ds.$$

That is, the value of the line integral with respect to arc length will remain the same, if we reverse the orientation of the curve.

But when we consider line integral with respect to x , then we are considering $f(x_i^*, y_i^*)\Delta x_i$. In case of line integral with respect to arc length, we were considering $f(x_i^*, y_i^*)\Delta s_i$. In whichever direction we consider, Δs_i is going to remain the same. It is a measure of the length of the subarc $P_{i-1}P_i$.

On the other hand,

$$\Delta x_i = x_i - x_{i-1} \text{ along } C$$

$$= x(t_i) - x(t_{i-1}).$$

And

$$\Delta x_i = x_{i-1} - x_i \text{ along } -C.$$

So the value of the line integral with respect to x (or y) will change sign if we consider the curve $-C$ instead of C .

1.6 Line integrals in space

This is exactly similar to line integrals in plane. The only differences are that the curve C is parametrized as

$$\vec{r}(t) = \langle x(t), y(t), z(t) \rangle, \quad a \leq t \leq b,$$

and

$$ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt.$$

And

$$\int_C f(x, y, z) ds = \int_a^b f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt.$$

1.7 Line integrals of vector fields

Earlier, when we computed the line integral $\int_C f(x, y, z) ds$ along a curve C in space, f was a scalar function of the 3 variables x, y, z . Now, instead of a scalar function f , we are going to consider a function

$$\vec{F}(x, y, z) = F_1(x, y, z)\hat{i} + F_2(x, y, z)\hat{j} + F_3(x, y, z)\hat{k},$$

which defines a vector field.

Earlier we had found that the work done by a constant force $\vec{F}$ in moving an object from a point P to another point Q in space is $W = \vec{F} \cdot \vec{D}$, where $\vec{D} = \overrightarrow{PQ}$ is the displacement vector.

Now suppose that $\vec{F}$ is a continuous force field on $\mathbb{R}^3$. We wish to compute the work done by this force in moving a particle along a smooth curve C .

We divide C into subarcs $P_{i-1}P_i$ with lengths Δs_i by dividing the parameter interval $[a, b]$ into subintervals. Choose a point $P_i^*(x_i^*, y_i^*, z_i^*)$ on the i -th

subarc corresponding to the parameter value t_i^* . If Δs_i is small, then as the particle moves from P_{i-1} to P_i along the curve, it proceeds approximately in the direction of $\vec{T}(t_i^*)$, the unit tangent vector at P_i^* . Thus the work done by the force $\vec{F}$ in moving the particle from P_{i-1} to P_i is approximately

$$\vec{F}(x_i^*, y_i^*, z_i^*) \cdot [\Delta s_i \vec{T}(t_i^*)] = [\vec{F}(x_i^*, y_i^*, z_i^*) \cdot \vec{T}(t_i^*)] \Delta s_i$$

and the total work done in moving the particle along C is approximately

$$\sum_{i=1}^n [\vec{F}(x_i^*, y_i^*, z_i^*) \cdot \vec{T}(x_i^*, y_i^*, z_i^*)] \Delta s_i \quad (*)$$

where $\vec{T}(x, y, z)$ is the unit tangent vector at the point (x, y, z) on C . Figure 1.7.1 illustrates the phenomenon.

Intuitively, we see that these approximations ought to become better as n

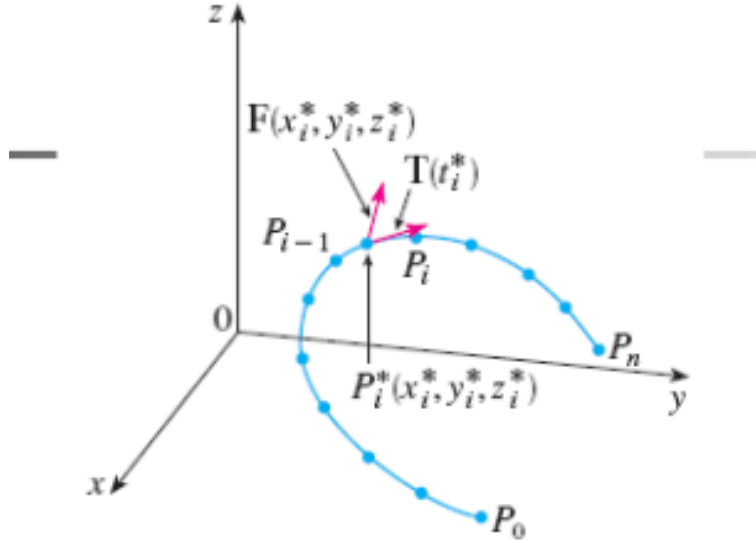


Figure 1.7.1: Line integrals of vector fields

becomes larger. Therefore, we define the **work** W done by the force field $\vec{F}$ as the limit of the Riemann sum in $(*)$, namely,

$$W = \int_C \vec{F}(x, y, z) \cdot \vec{T}(x, y, z) ds = \int_C \vec{F} \cdot \vec{T} ds.$$

The above equation says that work is the line integral with respect to arc length of the tangential component of the force.
 Now suppose $P(x, y, z)$ is any point on C and $\vec{OP} = \vec{r}(t) = \langle x(t), y(t), z(t) \rangle$, where O denotes the origin. Then we know that

$$\vec{T} = \frac{\vec{r}'(t)}{|\vec{r}'(t)|},$$

where $\vec{T}$ denotes the unit tangent vector to C at the point P . Also recall that

$$ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt.$$

Observe that

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} = |\vec{r}'(t)|.$$

So

$$ds = |\vec{r}'(t)| dt.$$

Then

$$\begin{aligned} \int_C \vec{F} \cdot \vec{T} ds &= \int_a^b \vec{F}(x(t), y(t), z(t)) \cdot \frac{\vec{r}'(t)}{|\vec{r}'(t)|} |\vec{r}'(t)| dt \\ &= \int_a^b \vec{F} \cdot \vec{r}'(t) dt. \end{aligned}$$

The last integral above is also written as $\int_C \vec{F} \cdot d\vec{r}$ because $d\vec{r} = \frac{d\vec{r}}{dt} dt = \vec{r}'(t) dt$.

Remark 1.7.1. If $\vec{F} = \langle F_1, F_2, F_3 \rangle$ and $\vec{r} = \langle x, y, z \rangle$, then

$$\vec{F} \cdot d\vec{r} = F_1 dx + F_2 dy + F_3 dz$$

and

$$\int_C \vec{F} \cdot d\vec{r} = \int_C F_1 dx + F_2 dy + F_3 dz,$$

where we have studied line integrals with respect to x and y . So we can work out line integrals with respect to z also in a similar way.

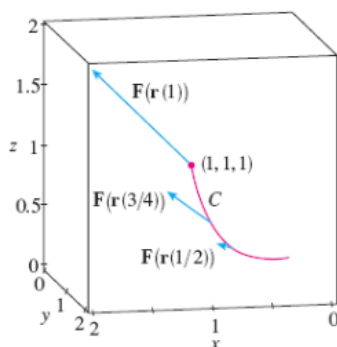


Figure 1.7.2: Figure for Example 2.7.2

Example 1.7.2. Let $\vec{F} = xy\hat{i} + yz\hat{j} + zx\hat{k}$ and C be the curve given by

$$x = t, \quad y = t^2, \quad z = t^3 : \quad 0 \leq t \leq 1.$$

Evaluate $\int_C \vec{F} \cdot d\vec{r}$. Figure 1.7.2 shows the curve C .

From Remark 2.7.1, we can write

$$\int_C \vec{F} \cdot d\vec{r} = \int_C F_1 dx + F_2 dy + F_3 dz,$$

where $F_1(x, y, z) = xy$, $F_2(x, y, z) = yz$, and $F_3(x, y, z) = zx$. So we have

$$\int_C F_1 dx + F_2 dy + F_3 dz = \int_0^1 t^3 dt + t^5(2t)dt + t^4(3t^2)dt = \int_0^1 (5t^6 + t^3)dt.$$

The rest of the calculation is left to the reader as an exercise. $\square$

Remark 1.7.3. In Example 2.7.2 above, an explicit parametrization of the curve C was given to us. But it may happen that sometimes an informal description of the curve C is given to you. Then you will have to find a parametrization of C .

1.8 The fundamental theorem for line integrals

The fundamental theorem of integral calculus can be written as

$$\int_a^b F'(x)dx = F(b) - F(a)$$

where F' is continuous on $[a, b]$.

Consider a curve C in a plane or in space whose initial point is A and final point is B . Figure 1.8.1 shows the curve C , where part (a) of the figure shows a planar curve and part (b) of the figure shows a space curve.

Then the **fundamental theorem** for line integrals states the following:

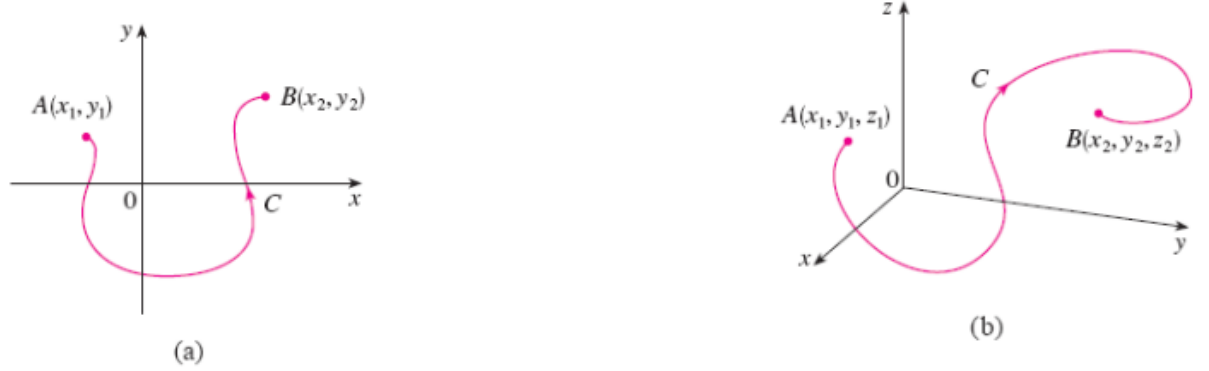


Figure 1.8.1: A curve C from the point A to the point B

Theorem 1.8.1. *Let C be a smooth curve given by the vector function $\vec{r}(t)$, $a \leq t \leq b$. Let f be a differentiable function of two or three variables whose gradient vector ∇f is continuous on C . Then*

$$\int_C \nabla f \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a)).$$

PROOF:

$$\begin{aligned} \int_C \nabla f \cdot d\vec{r} &= \int_{t=a}^b \nabla f \cdot \vec{r}'(t) dt \\ &= \int_a^b \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle \cdot \left\langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right\rangle dt \quad [\text{since } \vec{r}(t) = \langle x(t), y(t), z(t) \rangle] \\ &= \int_a^b \left(\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \right) dt \\ &= \int_a^b \frac{d}{dt} f(\vec{r}(t)) dt = [f(\vec{r}(t))]_{t=a}^b \end{aligned}$$

where the last equality follows from the fundamental theorem of integral calculus (from single variable calculus). The last expression above equals

$$f(\vec{r}(b)) - f(\vec{r}(a)).$$

This completes the proof. $\square$

Remark 1.8.2. The fundamental theorem says that if we perform the line integral of a vector field which is the gradient of a scalar function, then the value of the integral simply depends upon the values of the scalar function at the initial and the terminal points of the curve. That is, if we join the initial and the terminal points of the curve C by some other curve (lying in the same domain as C), then we are going to get the same value of the line integral. This is NOT true for an arbitrary vector field $\vec{F}$. This is true only when $\vec{F} = \nabla f$ for some scalar function f .

1.9 Conservative vector field

Definition 1.9.1. Consider a vector field $\vec{F}(x, y, z)$ in some domain $D \subseteq \mathbb{R}^3$. If $\vec{F}$ can be expressed as $\nabla\phi$ for some scalar function ϕ , then we call $\vec{F}$ a **conservative vector field**. The function ϕ is called the **potential function**. $\square$

1.10 Independence of path

Suppose C_1 and C_2 are two piecewise smooth curves (which are called **paths**) that have the same initial point A and terminal point B . In general, we have

$$\int_{C_1} \vec{F} \cdot d\vec{r} \neq \int_{C_2} \vec{F} \cdot d\vec{r}.$$

Please see Example 2.10.1 below in this regard.

Example 1.10.1. Evaluate the integral $\int_C y^2 dx + x dy$, where C is either C_1 or C_2 which are two curves as shown in Figure 1.10.1.

Observe that C_1 and C_2 have the same initial and terminal points. This example was discussed in one of the earlier lectures, where we had found that

$$\int_{C_1} y^2 dx + x dy \neq \int_{C_2} y^2 dx + x dy.$$

$\square$

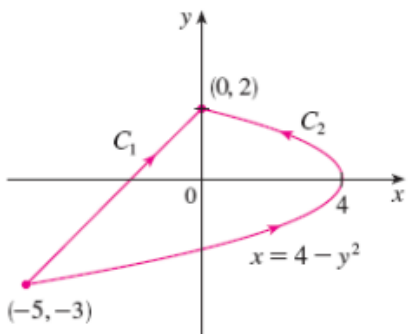


Figure 1.10.1: Figure for example 2.10.1

But the fundamental theorem for line integrals implies that

$$\int_{C_1} \nabla f \cdot d\vec{r} = \int_{C_2} \nabla f \cdot d\vec{r}$$

where ∇f is continuous. In other words, the line integral of a conservative vector field depends only on the initial point and terminal point of the curve.

Definition 1.10.2. If $\vec{F}$ is a continuous vector field with domain D , then we say that the line integral $\int_C \vec{F} \cdot d\vec{r}$ is **independent of path** if

$$\int_{C_1} \vec{F} \cdot d\vec{r} = \int_{C_2} \vec{F} \cdot d\vec{r}$$

for any two paths C_1 and C_2 in D that have the same initial points and the same terminal points. $\square$

With this terminology, we can say that the line integrals of conservative vector fields are independent of path.

Definition 1.10.3. Let C be a curve and $\vec{r}(t)$ be a parametrization of C where $a \leq t \leq b$. The curve C is called **closed** if its terminal point coincides with its initial point, that is, we have

$$\vec{r}(a) = \vec{r}(b).$$

$\square$

Theorem 1.10.4. $\int_C \vec{F} \cdot d\vec{r}$ is independent of path in a domain D if and only if $\int_C \vec{F} \cdot d\vec{r} = 0$ for every closed curve C in D .

PROOF: If $\int_C \vec{F} \cdot d\vec{r}$ is independent of path in D and C is any closed path in D , then we can choose any two distinct points A and B on C and regard C as being composed of the path C_1 from A to B followed by the path C_2 from B to A . Figure 1.10.2 shows a closed curve C and Figure 1.10.3 shows a breakup of the same closed curve C into C_1 and C_2 .

Then we have



Figure 1.10.2: A closed curve C

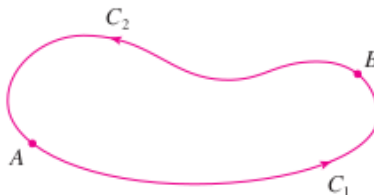


Figure 1.10.3: The curves C_1 and C_2

$$\int_C \vec{F} \cdot d\vec{r} = \int_{C_1} \vec{F} \cdot d\vec{r} + \int_{C_2} \vec{F} \cdot d\vec{r} = \int_{C_1} \vec{F} \cdot d\vec{r} - \int_{-C_2} \vec{F} \cdot d\vec{r} = 0,$$

where the last equality holds true since C_1 and $-C_2$ are two paths in D having the same initial and terminal points.

The converse part is left to the reader as an exercise. $\square$

1.11 Identifying conservative vector fields

For discussing this topic, we first need to understand the concepts of an open region, a connected region, and a simply-connected region. Let us first understand the concepts of an open region and a connected region. We will study simply-connected regions in the next lecture.

Definition 1.11.1. A region D in the plane is called **open** if for every point P in D , there exists a disk with center at P that lies entirely in D . The region D is called **connected** if any two points in D can be joined by a path that lies entirely in D . $\square$

Theorem 1.11.2. Suppose $\vec{F}$ is a vector field that is continuous on an open connected region D . If $\int_C \vec{F} \cdot d\vec{r}$ is independent of path in D , then $\vec{F}$ is a conservative vector field on D , that is, there exists a function f such that $\nabla f = \vec{F}$.

PROOF: Let (a, b) be a fixed point in D . We construct the desired potential function f by defining

$$f(x, y) = \int_{(a,b)}^{(x,y)} \vec{F} \cdot d\vec{r}$$

for any point (x, y) in D . Since $\int_C \vec{F} \cdot d\vec{r}$ is independent of path, it does not matter which path C from (a, b) to (x, y) is used to evaluate $f(x, y)$.

Since D is open, there exists a disk contained in D with center (x, y) . Choose any point (x_1, y) in the disk with $x_1 < x$ and let C consist of any path C_1 from (a, b) to (x_1, y) followed by the horizontal line segment C_2 from (x_1, y) to (x, y) . Please see Figure 1.11.1 for this path. Such a path exists because D is a connected region.

Then

$$f(x, y) = \int_{C_1} \vec{F} \cdot d\vec{r} + \int_{C_2} \vec{F} \cdot d\vec{r} = \int_{(a,b)}^{(x_1,y)} \vec{F} \cdot d\vec{r} + \int_{C_2} \vec{F} \cdot d\vec{r}.$$

Notice that the first of these integrals does not depend on x , so

$$\frac{\partial}{\partial x} f(x, y) = 0 + \frac{\partial}{\partial x} \int_{C_2} \vec{F} \cdot d\vec{r}.$$

If we write $\vec{F} = P\hat{i} + Q\hat{j}$, then

$$\int_{C_2} \vec{F} \cdot d\vec{r} = \int_{C_2} Pdx + Qdy.$$

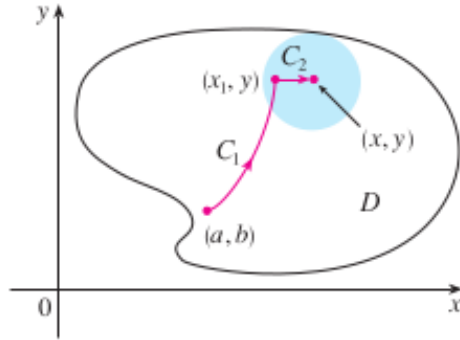


Figure 1.11.1: The curve C from (a, b) to (x, y)

On C_2 , y is constant, so $dy = 0$. Using t as a parameter, where $x_1 \leq t \leq x$, we have

$$\frac{\partial}{\partial x} f(x, y) = \frac{\partial}{\partial x} \int_{C_2} P dx + Q dy = \frac{\partial}{\partial x} \int_{x_1}^x P(t, y) dt = P(x, y),$$

where the last equality follows from the fact that

$$\frac{\partial}{\partial x} \left[\int_a^x P(\psi) d\psi \right] = P(x).$$

A similar argument, using a vertical line segment (see Figure 1.11.2) shows that

$$\frac{\partial}{\partial y} f(x, y) = Q(x, y).$$

Thus

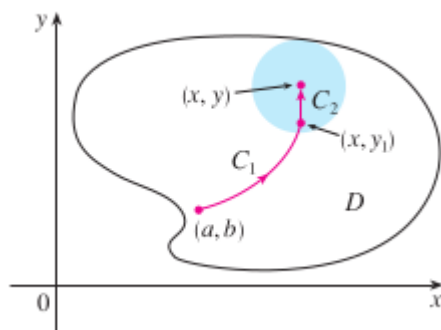


Figure 1.11.2: Another curve for the proof of Theorem 1.11.2

$$\vec{F} = P\hat{i} + Q\hat{j} = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} = \nabla f,$$

which says that $\vec{F}$ is conservative. □

—————-END—————